T-test

A sample of 10 students scored the following marks in an exam:

[72, 68, 75, 70, 74, 69, 71, 73, 70, 72] We want to test whether the **average mark = 70** ($\mu_0 = 70$) at **5% significance level** using python

Python Code:

```
import numpy as np

from scipy import stats

# Sample data

marks = np.array([72, 68, 75, 70, 74, 69, 71, 73, 70, 72])

# Hypothesized mean

mu_0 = 70

# One-sample t-test

t_stat, p_value = stats.ttest_1samp(marks, mu_0)

print(f"T-statistic: {t_stat:.3f}")

print(f"P-value: {p_value:.4f}")

alpha = 0.05

if p_value < alpha:

   print("Reject Null Hypothesis → Mean is significantly different from 70.")

else:

   print("Fail to Reject Null Hypothesis → No significant difference.")
```

OUTPUT:

```
T-statistic: 1.993
P-value: 0.0774
Fail to Reject Null Hypothesis → No significant difference.
```

Z-test

A manufacturer claims that the average weight of packets is **50 g**.A random sample of 36 packets has an average weight of **51.2 g** with a **known σ = 3 g**.At a **5% significance level**, test the claim.

Python Code:

```
import numpy as np

from math import sqrt

from scipy.stats import norm

# Given data

x_bar = 51.2   # sample mean

mu_0 = 50      # population mean

sigma = 3      # population standard deviation

n = 36         # sample size

# Calculate Z-statistic

z_stat = (x_bar - mu_0) / (sigma / sqrt(n))

# Two-tailed p-value

p_value = 2 * (1 - norm.cdf(abs(z_stat)))

print(f"Z-statistic: {z_stat:.3f}")

print(f"P-value: {p_value:.4f}")

alpha = 0.05

if p_value < alpha:

   print("Reject Null Hypothesis → Mean is significantly different from 50 g.")

else:

   print("Fail to Reject Null Hypothesis → No significant difference.")
```

Output:

```
Z-statistic: 2.400
P-value: 0.0164
Reject Null Hypothesis → Mean is significantly different from 50 g.
```

Anova Test

Three fertilizers (A, B, C) were tested on crop yield (in kg). Is there a significant difference among fertilizers? (Use $\alpha = 0.05$)

Fertilizer	Yields
A	20, 22, 23
B	19, 20, 18
C	25, 27, 26

Python Code:

```
import numpy as np

from scipy import stats


# Data

A = [20, 22, 23]

B = [19, 20, 18]

C = [25, 27, 26]


# Perform one-way ANOVA

f_stat, p_value = stats.f_oneway(A, B, C)


print(f"F-statistic: {f_stat:.3f}")

print(f"P-value: {p_value:.4f}")


alpha = 0.05

if p_value < alpha:

   print("Reject Null Hypothesis → Means are significantly different.")

else:

   print("Fail to Reject Null Hypothesis → No significant difference.")
```

OUTPUT

```
F-statistic: 25.923
P-value: 0.0011
Reject Null Hypothesis → Means are significantly different.
```